Journal of Environmental Chemical Engineering 13 (2025) 117766

Contents lists available at ScienceDirect

Journal of Environmental Chemical Engineering

journal homepage: www.elsevier.com/locate/jece

Lab to large-scale applications of microaeration in anaerobic digestion: Biowaste digestibility, microbiome nexus, and reactor stability

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ARTICLE INFO

Keywords:
Biogas
Biomethane
Desulfurization
Hydrogen sulfide
Microbial communities
Wastewater

ABSTRACT

Anaerobic digestion (AD) is improved with different approaches, including pretreatments, additives, co-digestion, and bioaugmentation. However, these techniques have limitations such as high cost, harmful substances production, and post-treatment of residues. Microaeration is preferable because it is cost-effective, eco-friendly, and easily operable. Recently, the effect of microaeration on AD has drawn researchers' attention. However, in-depth discussion about microbial communities' shifts, hydrogen sulfide (H_2S) removal, oxidative stress, antioxidative mechanisms, and biomethane efficiency under microaeration is still needed to be reviewed. Thus, the current review fully covered the recent research on the impact of microaeration on substrate digestibility, microbial diversity, desulfurization, and biomethane production. Growth of *Firmicutes* and *Bacteroidetes* is increased with microaeration, which facilitates the breakdown of organic compounds. The relative abundance of *Methanosaeta* is increased by 3–90.6 % with microaeration. Lab-scale studies exhibited biomethane enhancement by 7.8–77 % under microaeration. Several methanogens (such as *Methanosarcina barkeri* and *Methanobacterium thermoautotrophicum*) overexpressed antioxidative genes (2–30-fold) that encode enzymes such as catalase, superoxidase dismutase, and superoxide reductase. Large-scale microaeration in anaerobic wastewater treatment plants (WWTPs) removed $\geq$ 90 % H_2S. This review further suggests that integrating microaeration technology with bioaugmentation, co-digestion, and biochar could improve the reactor stability for efficient biomethanation and H_2S removal.

1. Introduction

Energy demand has increased due to the widespread consumption of natural energy resources (oil, natural gas, and coal) as a consequence of extensive population growth and industrial development [1]. Alternative sources of energy (solar, wind, and biofuel) can be utilized to fulfill the needs. Anaerobic digestion (AD) is one of the promising technologies for the generation of biofuels (biogas) from organic waste, including agricultural residues, municipal wastes, animal wastes, sewage sludge, wastewaters, and food wastes [2]. AD is a series of metabolic processes in which microorganisms break down organic waste materials without oxygen to produce biogas [3]. However, some limitations in AD must be overcome to improve the efficiency of the process. The key shortcoming of AD is the low hydrolysis rate. The organic wastes are hydrolyzed with very low frequency, and the process shows instability at a high organic loading rate (OLR) [4]. Metal pipes and biogas processing equipment are

List of abbreviations: AD, Anaerobic digestion; PS, Primary sludge; WAS, Waste activated sludge; WWTP, Wastewater treatment plant; COD, Chemical oxygen demand; VFAs, Volatile fatty acids; OLR, Organic loading rate; DO, Dissolved oxygen; Vmm, Air volume per gram TS per minute; ROS, Reactive oxygen species; SRB, Sulfate reducing bacteria; SOB, Sulfide oxidizing bacteria; SAOB, Syntrophic acetate oxidizing bacteria; ORP, Oxidation-reduction potential; OTR, Oxygen transfer rate; OUR, Oxygen utilization rate; SOD, Superoxide dismutase; SOR, Superoxide reductase; AHP, Alkyl hydroperoxide reductase.

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https://doi.org/10.1016/j.jece.2025.117766
Received 4 April 2025; Received in revised form 26 May 2025; Accepted 25 June 2025
Available online 26 June 2025
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also damaged by hydrogen sulfide (H_2S) and other compounds of sulfur (sulfur dioxides and sulfuric acid) produced during hydrolysis [5]. The utilization of wastes at high OLR leads to the accumulation of volatile fatty acids (VFAs), which decreases reactor pH and causes inhibition during acidogenesis [3,6].

AD has been improved with different methods, including physical (mechanical breakdown), chemical (acid and alkali), physicochemical (hydrothermal pretreatment), biological (enzymes), and co-digestion [7, 8]. Chemical pretreatment of lignocellulose biomass with acids and alkalis is used to improve substrate degradation rate, VFAs, and biogas production [9]. The additives (nutrients, nanomaterials, and biochar) are also used to enhance biogas production [10,11]. However, these approaches require more energy and chemicals, which not only increase operating expenses and complexity but also lead to several environmental issues [12]. For instance, biochar increases the total solid (TS) content, which results in a high fluid viscosity in the reactor. The rise in viscosity would reduce the heat and mass transfer in the reactor, which might slow the performance or results in failure of AD system [13]. It might also release heavy metals into the digestate, which would be a problem for the environment when used as fertilizers. More research is still needed to overcome slow digestibility, VFAs accumulation, and H_2S production during AD. Hence, the microaeration technique has been used for its efficient role in AD to improve biogas/biomethane production [14–16].

Microaeration refers to the addition of a very small quantity of air into AD to speed up hydrolysis, remove H_2S, increase VFAs production, and enhance biowaste utilization [14,17]. Since microaeration requires a small amount of air and can be easily operated on the spot in AD, it is more cost-effective and ecologically friendly than other methods [12, 16]. Current research trend of microaeration in AD determines its role in substrate digestibility, microbial communities' shift, and biomethane production. However, an in-depth investigation and discussion about the recent research related to the effect of microaeration on the overall AD process needs to be conducted. Furthermore, the integration of microaeration with other approaches (bioaugmentation, additives, and co-digestion) is rarely studied (see supplementary material).

Thus, the motivation of the current review is to collect and summarize the recent research data to fully assess the impact of microaeration on the hydrolysis rate, desulfurization, microbial diversity (bacteria and archaea), and biomethane yield. The protocol followed for the identification, screening, and selection of publications for the current review has been summarized (see supplementary material). The role of microaeration in desulfurization and biogas/biomethane enhancement has been explored. The approaches for the injection of air in AD and large-scale applications of microaeration in wastewater treatment plants (WWTPs) for the removal of H_2S have also been reviewed. Various integrated-microaeration strategies have been suggested to enhance the efficiency of AD. Further, challenges and future perspectives of microaeration in AD have been highlighted.

2. Anaerobic digestion and its limitations

Anaerobic digestion (AD) is a complex network of metabolic pathways facilitated by a vast community of anaerobes working in coordination to catalyze many biochemical reactions [18]. The dominant microbes in AD are *Firmicutes*, *Bacteroidota*, *Proteobacteria*, *Chloroflexi*, *Euryarchaeota,* and *Halobacteriota*, which play a vital role in AD to produce biogas (Table 1). Methane is the major constituent of biogas (50–70 %), including other components such as carbon dioxide (25–50 %), water vapor (0–10 %), hydrogen sulfide (0–3 %), and other trace gases (<1 %). A secondary byproduct of AD is a highly mineral-rich digestate [19].

The complex organic matter is broken down into simple molecules during hydrolysis [8]. Hydrolytic bacteria (*Micrococci*, *Bacteroides*, *Clostridia*, and *Streptococcus*) secrete hydrolases (amylases, proteases, and lipases) to degrade complex compounds [24]. Amylases help in the

Table 1
Microbial communities involved in different stages of AD reported in recent studies.

Phylum	Hydrolysis		Acidogenesis		Acetogenesis		Methanogenesis		Reference
	Genera	Facultative /obligate	Genera	Facultative /Obligate	Genera	Facultative /Obligate	Phylum	Genera	
Firmicutes	*Streptococcus*	Facultative	*Streptococcus*	Facultative	*Gallicola*	Obligate	*Euryarchaeota*	*Methanosarcina*	[20]
Bacteroidota	*Actinomyces*	Obligate	*Actinomyces*	Obligate	*Proteiniphilum*	Obligate	*Halobacteriota*	*Methanobacterium*	
Synergistota	*Fastidiosipila*	Obligate	*Fastidiosipila*	Obligate	*Syntrophaceticus*	Obligate		*Methanosphaera*	
	Aminobacterium	Obligate						*Methanoculleus*	
Firmicutes	*Proteiniphilum*	Obligate	*Proteiniphilum*	Obligate	*Longilinea*	Obligate	*Euryarchaeota*	*Methanosaeta*	[21]
Bacteroidota	*Defluviitoga*		*Defluviitoga*		*Syntrophomonas*	Obligate	*Halobacteriota*	*Methanosarcina*	
Proteobacteria	*Thermovirga*							*Methanobacterium*	
Chloroflexi								*Methanobrevibacter*	
								Methanolinea	
								Methanospirillum	
								Methanoculleus	
Firmicutes	*Fastidiosipila*	Obligate	*Gallicola*	Obligate	NR	NR	*Euryarchaeota*	*Methanosaeta*	[22]
Bacteroidota	*Gallicola*	Obligate					*Halobacteriota*	*Methanosarcinia*	
Cloacimonadota	*Bacteroidales_UCG−001*	Obligate						*Methanobrevibacter*	
								Methanospirillum	
Firmicutes	*Psychrobacter*	Facultative	NR	NR	NR	NR	*Euryarchaeota*	*Methanosaeta*	[23]
Proteobacteria							*Halobacteriota*	*Methanosarcina*	
Planctomycetes								*Methanobacterium*	
								Methanobrevibacter	
								Methanosphaera	
								Methanospirillum	
								Methanoculleus	

NR: Not reported

conversion of carbohydrates into simple sugars, while proteases break down proteins into amino acids. Lipases convert lipids into long-chain fatty acids (LCFAs) [25]. Acidogenic bacteria (*Clostridia*, *Lactobacilli*, propionic bacteria, and *Enterobacteria*) convert simple organic compounds into VFAs during acidogenesis [26]. VFAs are a group of organic acids, including smaller organic acids such as acetates and larger organic acids, including propionate and butyrate [27]. Organic acids produced in acidogenesis are converted into acetic acid during acetogenesis. A group of homoacetogenic bacteria (acetogens) carry out this process [28]. The end products of acetogenesis include acetate, hydrogen (H_2), and carbon dioxide (CO_2) (Fig. 1). The vital product of acidogenesis is acetic acid, which is utilized in methanogenesis during AD [29].

The acetic acid produced in acetogenesis is converted into methane by acetoclastic methanogens through acetoclastic methanogenesis [30]. Acetoclastic methanogens (*Methanosaeta* and *Methanosarcina*) are responsible for the majority of methane production [31,32]. Acetoclastic methanogenesis and syntrophic acetate oxidation are the two distinct processes that lead to the methanogenesis of acetate. Syntrophic acetate-oxidizing bacteria (SAOB) and acetoclastic methanogens are the competitors for acetate in this process [33]. The equation for acetoclastic methanogenesis [34] is as follows:

$$CH_3COOH \xrightarrow{\text{Acetoclastic methanogens}} CH_4 + CO_2$$

Hydrogenotrophic methanogens (*Methanobacterium*, *Methanoculleus*, and *Methanothermobacter*) play a crucial role in AD because they can allow syntrophic acetogens to continue to function while also consuming H_2 and CO_2 to generate methane [35]. Hydrogenotrophic methanogens are more resilient to environmental changes than acetoclastic methanogens [24]. The equation for hydrogenotrophic methanogenesis [36] is as follows:

$$CO_2 + 4H_2 \xrightarrow{\text{Hydrogenotrophic methanogens}} CH_4 + 2H_2O$$

The significant factors and parameters involved in AD are the types of substrate, microbial source, OLR, C/N ratio, pH, temperature, and oxidation-reduction potential (ORP) [37]. AD has certain limitations due to the complexity of the process. The major limitation of AD is the slow hydrolysis rate, which results in low substrate digestion [38]. It is important to enhance the hydrolysis rate for better substrate utilization. H_2S produced from sulfate by sulfate-reducing bacteria (SRB) through the dissimilatory sulfate reduction pathway negatively affects AD [39]. Sulfur-rich substrates result in the accumulation of sulfides in AD, which are toxic to methanogens and inhibit methanogenesis, leading to process failure [40]. The production of sulfuric acid corrodes the AD apparatus in large-scale reactors [5]. An effective approach is needed to overcome this difficulty in the AD system.

Low methane yield is mainly caused by the slow digestion of recalcitrant substrates [41]. Digesters are commonly run at high OLRs of feedstocks containing more total solids (TS) to overcome this limitation. However, excessive OLRs result in the buildup of VFAs, including butyric, acetic, and propionic acid, which causes AD failure [42]. This happens due to the imbalance between syntrophic acetogenic bacteria and methanogens [43]. It is imperative to design effective control systems to sustain the stability of the AD system. Microaeration in AD is suggested to increase the stability of the digester and, ultimately, biomethane production.

3. Microaeration

Microaeration involves the introduction of 0.005–5 L O_2/L reactor per day into AD reactors [15,44]. Other terminologies reported for this practice are micro-oxygenation, oxygenation, and microaerobic conditions [44]. Microaeration was initially used for desulphurization, which showed efficient elimination of H_2S from biogas [14,45]. However, it has been recently used to improve the efficiency and stability of the AD system to increase biomethane production [9,15,46]. The benefits of

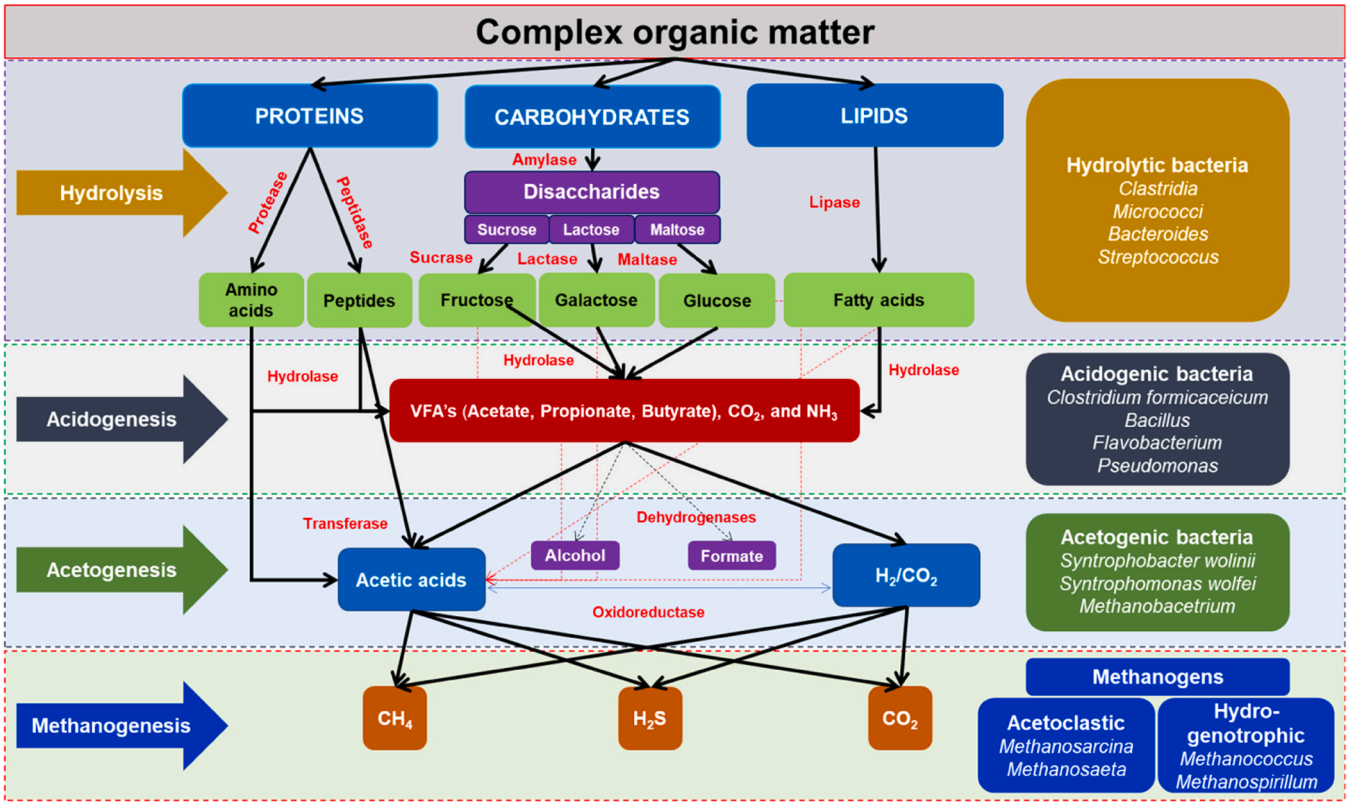


Fig. 1. An overview of AD focusing on each step comprising hydrolysis, acidogenesis, acetogenesis, and methanogenesis. The major enzymes and microbes involved in each step are mentioned.

microaeration in AD are further discussed.

3.1. Role of microaeration

3.1.1. Hydrolysis

Hydrolysis is typically regarded as the rate-limiting phase, particularly in the case of resilient biomasses [47]. Hydrolysis is performed by extracellular hydrolytic enzymes (amylases, proteases, lipases, and cellulases), released by facultative bacteria. Chemical, mechanical, biological, and thermal pretreatments have been used to speed up hydrolysis [48]. However, the widespread use of these procedures is constrained by economic and environmental factors. A practical alternative, in this case, would be microaeration or microaerobic pretreatment in the AD. It has been observed that the richness and diversity of hydrolytic bacteria can be promoted with microaeration [49]. Consequently, the number of extracellular hydrolytic enzymes will increase, ultimately leading to enhanced hydrolysis of biowastes [44,50]. Facultative microbes and obligate anaerobic bacteria may cohabit in such a way that it would allow the consumption of oxygen traces by facultative microbes and maintain an anaerobic environment for strict anaerobes. Thus, microaeration is suggested to strengthen the hydrolysis rate by promoting the growth of hydrolytic microbes in AD (Fig. 2).

3.1.2. Microbial community distribution

Microorganisms in AD interact with each other in such a way that there is a continuous transfer of byproducts among different stages. Bacterial and archaeal communities can carry out their essential functions with mutual dependence. To ensure a consistent operation of the microaerobic AD process, equilibrium and syntrophism among various microbes should be preserved. Microbial communities in AD are significantly affected by microaeration (Table 2). The microbial communities are developed differently according to substrate types [51], inoculum types [52], and operational parameters (such as pH and temperature) [53]. The growth of microbial populations involved in hydrolysis and fermentation is encouraged by microaeration [54,55]. The abundance of *Firmicutes* increased from 58 % to 72 % due to microaeration, which increased hydrolysis of food waste and brown water [55]. According to a metaproteomic study, the enzymatic degradation of cellulose and other polymers is carried out by *Bacteroidetes* and *Firmicutes* [51,56].

The Archaea domain typically comprises 4 % of the microbial population; however, metagenomic research revealed that this is responsible for 30 % of the protein produced during AD [51]. The primary methane-generating archaea are acetoclastic methanogens (AM), including *Methanosaeta, Methanosarcina, Methanoculleus, Methanobacteriales*, and *Methanospirillum* [59,60]. Hydrogenotrophic methanogenesis is the predominant pathway in the thermophilic AD process with high concentrations of VFAs and ammonia [61]. This might be because AM is less resistant to inhibitors as compared to syntrophic acetate oxidizing bacteria (SAOB) and hydrogenotrophic methanogens (HM) [62,63]. When hydrolysis and acidogenesis are improved in a microaerobic environment, more precursors are produced for methanogens, which leads to increased methanogenic activity and methane output [54,59]. *Methanosarcina* is considered the dominant archaea in AD under microaeration, which is aerotolerant and can produce biomethane while using both acetate and hydrogen [54]. The major archaeal genera shifted from *Methanosarcina* to *Methanoculleus* under microaeration, resulting in a change of the metabolic route from the acetoclastic to the hydrogenotrophic pathway [60,64].

The variation in bacterial and archaeal communities upon the application of different microaeration rates in various substrates are reported in previous studies (Fig. 3). The abundance of *Chloroflexi* and *Firmicutes,* responsible for substrate degradation was increased after application of 12.5 mL/L per day microaeration to mixed sludge (PS:40 and WAS:60 %) in a continuously stirred tank reactor [12]. *Methanosaeta* involved in acetoclastic methanogenesis was also enhanced, while a repression was reported for hydrogenotrophic *Methanobacterium* (Figs. 3-a2). In another study, 4 vmm (air volume per gram TS per minute) microaeration was injected in anaerobic sludge (PS: 60 % and WAS: 40 %), which increased the abundance of *Proteobacteria*, *Chloroflexi*, and *Synergistetes* (Figs. 3-b1). *Proteobacteria* and *Chloroflexi* help in hydrolysis and acidogenesis, and *Synergistetes* speed up acetogenesis, thus providing more acetate for methanogenesis to increase biomethane production [57]. The abundance of *Methanosaeta* was higher than that of *Methanobacterium* with 4 vmm microaeration in the mixed sludge (Figs. 3-b2), which showed that acetoclastic methanogenesis was the main pathway for biomethane production.

Microaeration dosage of 5 mg/g COD in blackwater enhanced the

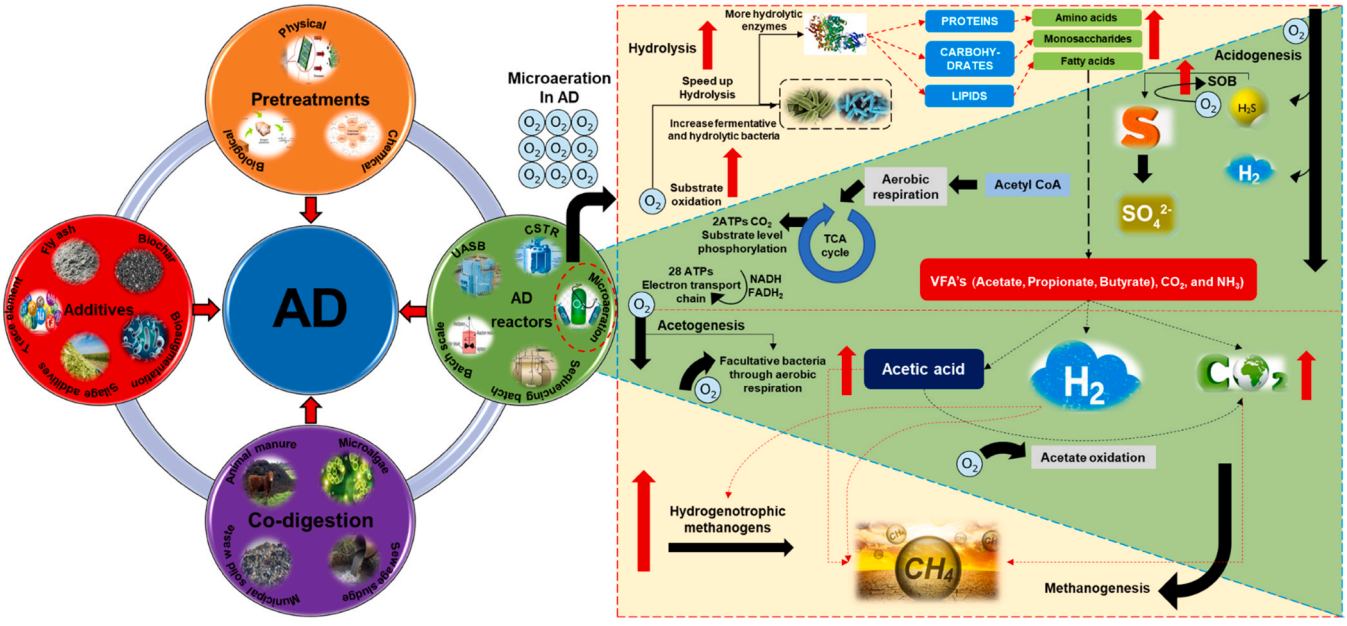


Fig. 2. Various strategies used for AD enhancement, including pretreatments, additives, co-digestion, and microaeration. The role of microaeration in AD is briefly summarized.

Table 2
Previous studies show the effects of microaeration on microbial communities in AD using different substrates and microaeration rates.

Substrate	Reactor configuration					Microaeration rate	Archaeal communities		Bacterial communities		References
	Reactor type	Total volume	Working volume	Temperature condition	AD time (days)		Enhanced	Inhibited	Enhanced	Inhibited	
PS (40 %) & WAS (60 %)	Continuous stirred-tank reactors (CSTR)	NR	14 L	Mesophilic	40	12.5 mL/L per day	*Methanosaeta* *Methanomicrobia*	*Methanobacterium*	*Bacteroidetes* *Chloroflexi* *Actinobacteria* *Acidobacteria* *Firmicutes*	*Proteobacteria* *Synergistetes*	[12]
PS (60 %) & WAS (40 %)	Batch-scale	250 mL	200 mL	Mesophilic	12	4 vmm	*Methanosaeta* *Methanobacterium* *Methanolinea*	*Methanobrevibacter* *Methanospirillum*	*Proteobacteria* *Chloroflexi* *Actinobacteria* *Synergistetes*	*Firmicutes* *Parcubacteria*	[57]
Corn straw	Batch-scale	300 mL	200 mL	Mesophilic	40	0.2 mL/(g·VS) per day	*Methanosaeta* *Methanobacterium*	*Methanosarcina*	*Firmicutes* *Synergistetes* *Spirochaetes*	*Bacteroidetes*	[46]
Black water	Sequencing batch	1000 mL	710 mL	Mesophilic	35	5 mg/g COD_{feed}	*Methanosaeta* *Methanosarcina* *Methanobacterium* *Methanospirillum*	*Methanolinea*	*Proteobacteria* *Bacteroidetes* *Firmicutes*	*Chloroflexi* *Actinobacteria*	[58]
Corn straw	Batch-scale	300 mL	200 mL	Thermophilic	30	12.5 mL/L per day	*Methanosarcina* *Methanobacterium*	*Methanosaeta* *Methanoculleus* *Methanothermobacter*	*Firmicutes*	*Chloroflexi* *Synergestetes*	[54]

AD: Anaerobic digestion; PS: Primary sludge; WAS: Waste-activated sludge; vmm: Air volume per gram TS per minute; NR: Not reported.

abundance of *Bacteroidetes* (Figs. 3-c1), followed by *Proteobacteria* and *Firmicutes* [58]. The archaeal community showed a high abundance of *Methanosaeta* compared to *Methanobacterium* (Figs. 3-c2). A microaeration dose of 12.5 mL/L per day was injected in the thermophilic AD of corn straw, which increased the abundance of *Firmicutes* (Figs. 3-d1), which are common bacteria that produce extracellular enzymes (lipase, protease, and cellulase). The high abundance of these bacteria under microaeration increased the metabolism of lipids, protein, and cellulose, thus enhanced the degradation of corn straw [54]. The relative abundance of oxygen-tolerant methanogens (*Methanobacterium* and *Methanosarcina*) was also increased (Figs. 3-d2), ensuring efficient and stable methanogenesis under microaerobic conditions.

3.1.3. Scavenging hydrogen sulfide

Hydrogen sulfide (H_2S) is produced via the sulfate reduction of sulfur-rich biomass by sulfate-reducing bacteria (SRB), including *Desulfobacterium*, *Desulfovibrio*, and *Desulfomicrobium*. H_2S inhibits obligate anaerobes, particularly methanogens, and is corrosive to the reactor pipes and metal parts. It also harms gas motors under various applications, including power generation, vehicles, and household usage supply chains. Thus, it is vital to eliminate H_2S from biogas in energy generation and transportation. Traditional methods for removing H_2S from biogas are based on physicochemical techniques (washing with alkaline and filtration with iron oxide filters) [65]. In biological methods, sulfide-oxidizing bacteria (SOB) such as *Thiobacillus*, *Halothiobacillus*, *Sulfuricurvum*, and *Acinetobacter* are used for the oxidation of H_2S to elemental sulfur (S). However, in this case, biogas cleaning typically takes place in special desulfurization units.

Direct desulfurization in biogas reactors with microaeration has recently gained attention. Sulfides are oxidized to elemental sulfur by SOB due to microaeration within the liquid phase of the AD reactor or by injecting air into the headspace [66]. Facultative chemolithotrophs can grow either chemolithoautotrophically with an inorganic energy source and CO_2 as a carbon source, or heterotrophically with organic compounds as a source of carbon and energy. Some *Thiobacilli* sp., certain *Beggiatoa*, *Thiosphaera pantotropha*, and *Paracoccus denitrificans* are typical examples of facultative chemolithotrophic SOB. Chemolithoheterotrophs such as *Thiobacillus* sp., and some *Beggiatoa* strains generate energy from the oxidation of reduced sulfur compounds under microaeration [66].

Sulfide oxidation occurs at the gas-liquid interface or on the walls of the headspace of the reactor for in-situ H_2S elimination under microaeration [67]. This is frequently carried out for biowastes containing high concentrations of sulfides, such as livestock and sewage sludge wastes, achieving notable advancements in H_2S removal. SOB catalyze the sulfide oxidation processes by generating the Sox protein complex and are responsible for the biological oxidation of sulfides (S^{2-}/HS^-) to elemental sulfur (S°), sulfate (SO_4^{2-}), and thiosulfate ($S_2O_3^{2-}$) [68]. The main genera of SOB (*Halothiobacillus, Thiobacillus, Acinetobacter*, and *Sulfuricurvum*) belong to the *Proteobacteria* [69]. High activity of SOB removed 74–99 % of H_2S from municipal wastewater with the application of a microaeration dose of 0.28–6 h in WWTPs in Central Europe. Significant reduction in soluble chemical oxygen demand (SCOD) was observed from 18 % to 33 % [70]. In another study, 30 mL/g VS microaeration for 15 days during the digestion of chicken manure increased *Proteobacteria* by 32 %, comprising the majority of SOB, resulting in 58 % H_2S removal [71].

The microaeration rate for sulfide removal can be determined by utilizing the molar ratio of O_2: S^{2-}. Different molar ratios of O_2: S^{2-} ranging from 0.15 to 2.0, have been documented in the literature [69]. According to stoichiometry, the reaction requires 0.5 mol of O_2 for every mol of S^{2-} to transform H_2S into S. However, a number of variables, including sulfide concentration, soluble substrate concentration, and pH, among others, have an impact on rate of sulfide oxidation. Dissolved sulfides found in biowastes and wastewater, and H_2S in biogas might both be eliminated by microaeration in liquid and gaseous phase

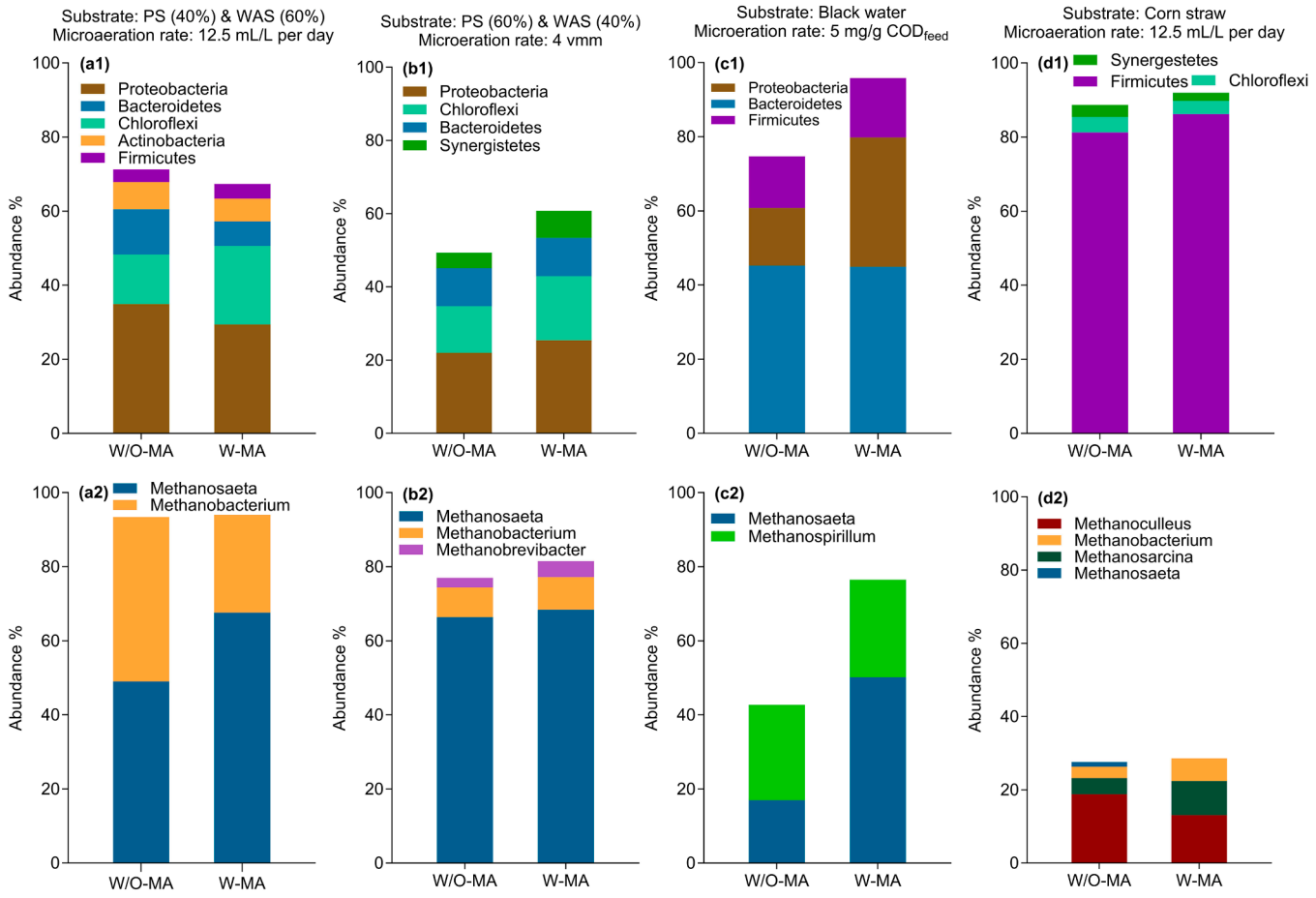


Fig. 3. The relative abundance of microbial communities with microaeration (W-MA) and without microaeration (W/O-MA) reported in previous studies [12,54,57,58]. The relative abundance of archaea is shown in **fig. a1, b1, c1**, and **d1,** while the abundance of bacteria is shown in **a2, b2, c2**, and **d2**. The most common archaea are *methanosaeta, methanobacterium, methanosarcina*, and *methanospirillum*. The major bacteria are *firmicutes, bacteroidetes, proteobacteria*, and *chloroflexi*. PS: primary sludge; WAS: waste activated sludge; vmm: air volume per gram TS per minute; COD_{feed}: substrate fed per cycle.

Table 3
Reported studies from the research of the last 6 years on the use of different microaeration rates with various substrates for enhancing biogas/biomethane production.

Substrate	Microbial source	Reactor configuration					Microaeration		Enhancement		References
		Reactor type	Total volume	Working volume	Temperature condition	AD Time (days)	O_2/air dose	Time of O_2/air supply	Biomethane	Biogas	
Mix sludge (60:40)	Sludge from WWTP	Batch-scale	250 mL	200 mL	Mesophilic	12	0–6 vmm	At the beginning	21.6 % at 4 vmm	16.4 % at 4 vmm	[57]
Corn straw	Anaerobic sludge	Batch reactor	300 mL	200 mL	Mesophilic	40	0–1.6 mL/(gVSday)	Per day	7.8 % at 0.2 mL/(gVSday)	NR	[46]
Dry corn stover	Fresh cow manure	Batch scale	500 mL	NR	Mesophilic	20	0.1–1 mg/L DO	At the beginning	16.6 % at 5 mg/L DO	NR	[77]
Chicken manure	Anaerobic sludge	Batch reactor	120 mL	70 mL	Mesophilic	35	7–50 mL/g VS_{in}	First 15 days	77 % at 30 mL/g VS_{in}	NR	[45]
Black water	Anaerobic and aerobic-activated sludge	Sequencing batch	1000 mL	710 mL	Mesophilic	35	0–150 mg/g COD_{feed}	At the beginning	NR	50.9 % at 5 mg/g COD_{feed}	[58]
Buffalo manure	Anaerobic sludge	Batch reactor	1000 mL	600 mL	Mesophilic	15	Phase 1: 7.3 mL/g VS Phase 2: 5.3–9.3 mL/g VS	Phase 1: at 6–47 h Phase 2: at 12, 24 h	NR	32 % average daily biogas at 7.3 mL O_2/g VS	[16]

AD: Anaerobic digestion; WWTP: Wastewater treatment plant; NR: Not reported; vmm: Air volume per gram TS per minute; DO: Dissolved oxygen; VS: Volatile solids.

respectively [72,73].

SRB competes with methanogens for acetate and CO_2, which results in the accumulation of H_2S in AD [74], resulting in low biogas/biomethane yield [75,76]. The competition between SRB and methanogens for precursors (acetate, H_2, and intermediates) must be considered to achieve high biomethane production. Hence, microaeration in AD is suggested to decrease the growth of SRB for high biogas/biomethane yield.

3.1.4. Enhancement of biogas/biomethane

The primary goal of microaeration in AD is to increase biogas/biomethane yield. This can be achieved by providing the optimum dose of microaeration at a specific time. Different units are adapted for adding air in AD, such as vmm (air volume per gram TS per minute), mL/gVSday (air volume per gram VS per day), mg/L DO (dissolved oxygen per L of working volume), and mg/g COD_{feed} (oxygen dosage per g COD feed). Previous studies have shown a beneficial effect of microaeration on biogas/biomethane production (Table 3). It is reported that a microaeration dose from 0 to 6 vmm was used in AD, which increased biomethane and biogas production by 21.6 % and 16.4 %, respectively, at 4 vmm [57]. Microaeration enhanced *Proteobacteria*, *Chloroflexi*, *Synergistetes*, and *Actinobacteria* by 15.3 %, 37.9 %, 77.7 %, and 58.3 %, respectively, which were associated with the degradation of sludges to increase SCOD by 33.6 % [57].

Microbial pretreatment with 5 mg/L DO at pH 8 facilitated the degradation of corn stover, and elevated the SCOD content. The high SCOD concentration resulted in increased VFAs production (13.22 g/L) with the acetic acid concentration upto 86 %, which enhanced biomethane yield by 16.6 % [77]. *Bacteroidetes* and *Firmicutes* were increased by 44.4 % and 16 %, respectively, after application of 5 mg/g COD, resulting in 16.6–53.1 % increased hydrolysis of blackwater. Methanogenesis also improved from 15.7 % to 50.9 % due to the increased abundance of *Methanosaeta* from 17.0 % to 50.2 % [58]. In another study, microaeration was supplied for the first 15 days of the process at a rate of 7–50 mL/g VS, which increased biomethane production by 77 % at 30 mL/g VS [45]. The relative abundance of *Methanocelleus* was increased, which favored the hydrogenotrophic methanogenesis. Intermittent microaeration (0–1.6 mL/g VS day) was used in a study, which enhanced biomethane yield by 7.8 % at 0.2 mL/g VS day [46]. The high cellulase activity after microaeration enhanced the degradation of corn straw, which increased the VFAs production. It is concluded from the previous research that microaeration effectively enhances biomethanation when it is provided in an optimum range. A very low microaeration dose has almost no effect on biomethane production, while a high dosage inhibits the process.

3.2. Microaeration-based AD control and oxidation-reduction potential (ORP)

The performance of a microaeration-based AD is influenced by the microaeration rate, the technique of aeration, the nature of the inoculum and substrate, and other parameters (pH and temperature). Optimization of variables is crucial for enhancing the stabilization and effectiveness of AD [44]. The rate of microaeration is the most critical factor and should be provided in a controlled manner in AD to enhance efficiency and stability. The oxygen utilization rate (OUR) and oxygen transfer rate (OTR) must be accounted for effective microaeration [14]. The OTR is impacted by several factors, including reactor design, dosing technique, and the TS of feedstock [78]. The OUR relies on the kind of substrate and inoculum used in AD [3]. Oxidation-reduction potential (ORP) is a controlling parameter for effective microaeration rate and keeping the balance between OTR and OUR.

ORP is the measure of the propensity of an aqueous solution to donate or accept electrons [14,44], which is frequently used to analyze the oxidative and reductive states of the solution. ORP is a regulatory parameter to sustain appropriate oxygen in the AD reactor without affecting obligate anaerobes. ORP fluctuates consistently with the amount of oxygen in the aqueous solution [79,80]. A microaerobic state is considered an aqueous solution with ORP between 0 and −300 mV as compared with the standard hydrogen reference electrode (SHE) [69,72]. The ORP of an anaerobic condition is lower than −300 mV, while the ORP of an aerobic environment is from 0 mV to + 820 mV (Table 4). Facultative microbes can be cultivated in both aerobic and anaerobic conditions [81,82]. However, these microorganisms show optimum growth within ORP between 0 and −300 mV and produce fermented products such as ethanol, butanol, lactic acids, and other VFAs. Furthermore, the most common redox pairs in AD are CO_2/CH_4 and SO_4^{2-}/HS^-, which have ORP in the microaerobic range. The value of ORP becomes more positive with the injection of air in AD and then gradually falls when facultative microbes utilize the oxygen [3].

Some researchers have used ORP (-100 to −200 mV) as a controlled factor for VFAs production under microaeration [52]. Another study used ORP between −275 and −265 mV for the removal of sulfides [84]. The aqueous and gaseous sulfides were entirely removed during microaeration at ORP between −230 and −180 mV, and the methane production was enhanced by 56.3 % [72,85,86]. The ORP was increased from −320 to −270 mV after six days of microoxygenation, resulting in a reduction of H_2S concentration in biogas from 6000 to 30 ppm [79]. These investigations showed the dependency of attaining sulfide removal by ORP-controlled microaeration. Therefore, ORP can be used as a controlled parameter for microaeration to maintain reactor stability and achieve high biogas yield.

Table 4
Oxidation-reduction potential (ORP) is a parameter for different metabolic processes. The data in the table is adapted from [3,14,44,83].

ORP (mV)	Representing environment	Redox pairs	Metabolic process	Microbes involved
+ 820	Aerobic	½ O_2/H_2O	Aerobic respiration	Aerobic and facultative bacteria
+ 750	Aerobic	$NO_3^-/½\ N_2$	Denitrification	Facultative bacteria
+ 360	Aerobic	NO_3^-/NH_4^+	Dissimilatory nitrate reduction	Facultative bacteria
+ 170	Aerobic	$HS^-/S°$	Sulfide oxidation	Facultative sulfide oxidizing bacteria
−57	Microaerobic	Rubredoxin Ox/Red	Anti-oxidation	Facultative bacteria and archaea
−100	Microaerobic	NR	Xylitol fermentation	*Candida tropicalis*
−150	Microaerobic	NR	Ethanol fermentation	*Saccharomyces cerevisiae*
−190	Microaerobic	NR	1,3-propaneiol fermentation	*Klebsiella pneumoniae*
−190	Microaerobic	Pyruvate/ lactate	Lactic acid fermentation	Facultative bacteria
−220	Microaerobic	SO_4^{2-}/HS^-	Sulfate reduction	Anaerobic sulfate-reducing bacteria
−240	Microaerobic	CO_2/CH_4	Methanogenesis	Anaerobic methanogens
−280	Microaerobic	CO_2/acetate	Acetogenesis	Anaerobic bacteria
−290	Microaerobic	NR	Butanol fermentation	*Clostridium acetobutylicum*
−320	Anaerobic	NAD^+/ NADH	Energy production, electron transfer	Bacteria and archaea
−350	Anaerobic	NR	Succinic acid fermentation	*Actinobacillus succinogenes*
−420	Anaerobic	2 H^+/H_2	Anaerobic reduction	Anaerobic and facultative bacteria

ORP: Oxidation-reduction potential; mV: Millivolt; NR: Not reported.

3.3. Oxidative stress and antioxidative mechanisms

Oxidative stress is a disturbance in the prooxidant-antioxidant balance in favor of the former [87]. This is the stress produced by reactive oxygen species (ROS), which adversely affect obligate anaerobes (Fig. 4a). Extreme oxidative stress is harmful to microbes, as the accumulation of ROS in the cells affects proteins (inactivation), lipids (cell membrane damage), biomolecules (enzyme destruction), and nucleic acids (breaking DNA) [88]. ROS are the reactive species of oxygen generated as a result of the partial reduction of oxygen molecules by facultative anaerobes in a microaerobic environment. The major types of ROS are O_2^-, H_2O_2, and •OH. These reactive ions and radicals of oxygen are severely detrimental to obligate anaerobes [89]. Aerobes and facultative anaerobes typically have complex antioxidant defense mechanisms to overcome ROS and diminish their effects. These mechanisms frequently involve the activation of specialized redox-sensitive transcriptional regulators, an increase in the expression of genes encoding scavenging enzymes, and the action of cellular repair mechanisms [88,90].

Obligate anaerobes utilize respiratory electron transport chains in which oxygen is not the terminal electron acceptor [91]. It was previously believed that obligate anaerobes are unable to combat and survive oxidative stress due to a lack of a functional ROS defense system [92]. However, recent research showed that certain obligate anaerobes can also synthesize antioxidative enzymes to adapt to oxidative conditions under microaeration. Some obligate anaerobes (*Clostridium acetobutylicum*, *Clostridium perfringens*, *Desulfovibrio gigas*, *Bacteroides fragilis*, *Methanobrevibacter*, *Methanobacterium*, and *Methanosarcina barkeri*) overexpress the genes (2–30 folds) that encode antioxidative enzymes such as catalase, superoxide dismutase (SOD), and superoxide reductase (SOR) under microaeration [93,94]. These anaerobes showed survival after being exposed to H_2O_2 (500 μM), O_2 (5 % v/v) in the headspace, or DO (5.6 mg/L) under microaeration for 1 h [94].

Obligate anaerobes have some antioxidative mechanisms against oxidative stress under microaerobic conditions (Figs. 4b and 4c). The first mechanism is the formation of flocs, aggregates, or biofilms by different microbial groups in the AD reactor. During floc formation, obligate anaerobes are surrounded by facultative anaerobes to protect them from oxidative stress (Fig. 4b). Thus, the obligate anaerobes are protected from O_2 due to the low diffusion of O_2 and its fast consumption by facultative anaerobes through antioxidative enzymes [14,95]. The second mechanism is the production of antioxidative enzymes by some methanogens, such *as Methanosarcina, Methanobacterium*, and *Methanocella* [96,97]. The most common antioxidative enzymes produced by these microbes are SOR, rubredoxin oxidoreductase, (SOD), alkyl hydroperoxide reductase (AHP), and catalase. These enzymes act on ROS and detoxify them to produce water (Fig. 4c). However, over-aeration in AD would inhibit the survival of obligate anaerobes as the facultative anaerobes will be unable to utilize a high amount of oxygen, and ultimately result in the inhibition of AD. Therefore, it is necessary to have controlled microaeration in AD to achieve maximum biogas/biomethane yield.

3.4. Current approaches for the injection of air

Different dosing methods have been adapted for microaeration, such as pretreatment with microaeration, intermittent supply, and continuous addition of air (Fig. 5). During pretreatment, air is directly supplied into the substrate or sludge before the main AD experiment. This method is adapted to enhance the reactor performance by promoting the growth of facultative microbes. When microaeration is used as an initial treatment method, paying close attention to the aeration frequency is essential. High oxygen concentrations can prevent the growth of methanogens, which lowers methane production, while less oxygen may

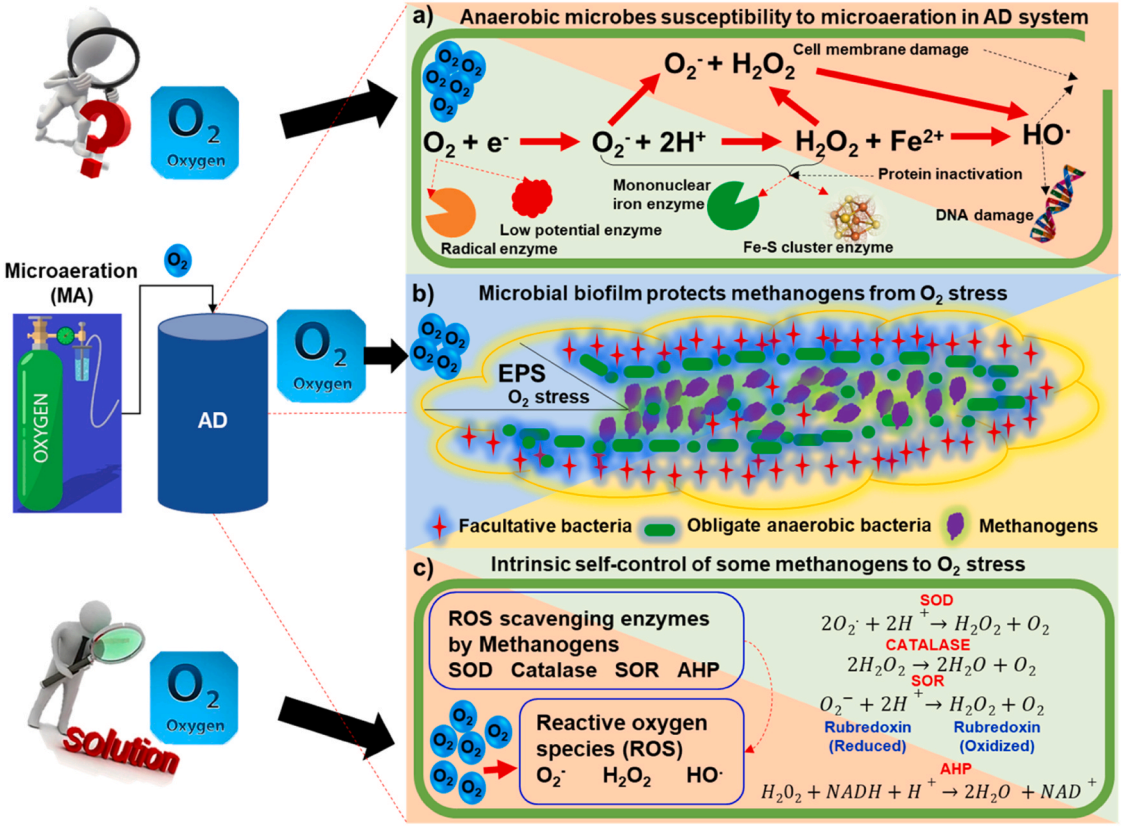


Fig. 4. The addition of oxygen in AD affects methanogens and obligate anaerobic bacteria. **a)** The susceptibility of obligate anaerobes to oxidative stress is due to the production of reactive oxygen species (ROS). **b)** and **c)** show the survival mechanisms of some methanogens against oxidative stress.

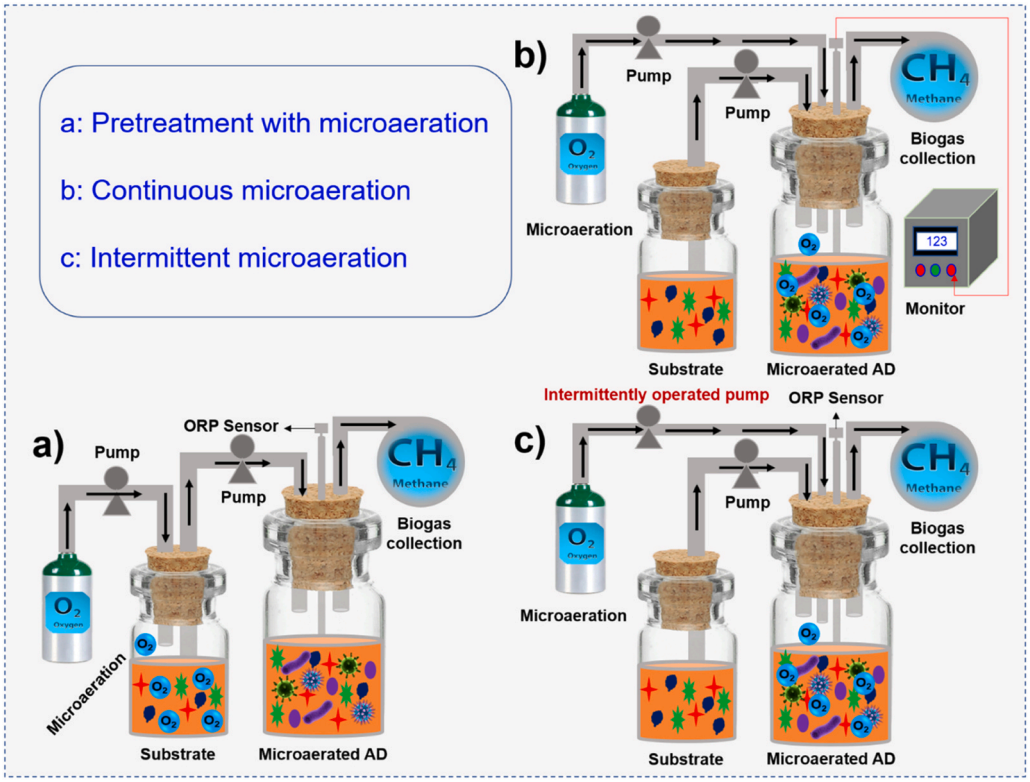


Fig. 5. Dosing approaches adapted for microaeration in AD.

have no effect. Therefore, the optimum microaeration dose should be used for high biogas generation. Some studies have reported pretreatment with microaeration to speed up hydrolysis. For instance, the microaeration pretreatment of sludge for four hours facilitated the growth of hydrolytic and fermentative bacteria to enhance hydrolysis and acidification, resulting in an increase in biomethane yield by 21.6 % [57]. A microaerobic pretreatment for 12 h was reported in another study to enhance the efficiency of AD on lignocellulosic substrate, which showed an increase in biomethane yield of 16.6 % [77]. However, the facultative microbes completely utilize the oxygen supplied during pretreatment, and no more oxygen will be available during the main AD experiment. Thus, an intermittent microaeration strategy is implemented to keep the oxygen concentration at a sufficient level to maintain the microaerobic environment in the AD reactor. A microaeration dose of 37.5 mL/(L min) per day was reported to increase hydrolytic enzymes, promote acetoclastic methanogenesis, which enhanced biomethane yield by 13.1 % [98]. In another study, 5 mg O_2/L reactor was injected for 5 min during each cycle (after 2–5 days) for 100 days, which increased hydrolysis of blackwater from 34.7 % to 48.7 %. Moreover, the biomethane production was enhanced from 39.6 % to 50.7 % [99]. A continuous and regulated supply of oxygen can achieve the ideal microaerobic condition. Daily microaeration (12 N mL air/day) decreased the lag phase of co-digestion of poultry litter and wheat straw by 3.4 % and increased the cumulative biomethane yield by 16.5 % [15]. In another study, biomethane yield was enhanced by 7.8 % at a daily microaeration dose of 0.2 mL/g VS per day while promoting the abundance of *Methanobacterium* [46].

The different techniques employed for injecting air into AD, such as the use of a syringe, natural aeration, and a peristaltic pump [3], are summarized (Table 5). Natural aeration is done in open air as a microaerobic pretreatment, as observed in a previous study for enhancing the degradation of lignocellulosic substrate [77]. However, this method needs an additional detector (DO detector) for controlling the DO of the reactor, and there are more chances of contamination with air microbes and dust particles. In another study, a peristaltic pump was used for controlled microaeration with an air flow rate of 200 mL/min for increasing VFAs and lactic acid production in agricultural wastes. The VFAs were enhanced by 20.3 %, and lactic acid production by the Bifidum pathway was increased by 22.9 % [9]. The advantage of this method is controlled microaeration with an accurate flow rate; however, this technique has additional costs for installing extra equipment. Therefore, syringes are widely used for microaeration due to their low cost and ease of availability everywhere. The rate of microaeration (microaeration intensity) in each technique is a very important factor for exploring the effects of microaeration in AD [44].

3.5. Large-scale applications of microaeration

The applications of microaeration on a large scale are limited and mostly focused on the removal of H_2S from AD. Most of the studies used microaeration for the removal of H_2S from the AD reactor. The commonly used substrates were sludge, wastewater, and cow manure in large-scale applications (Table 6). Most of the reactors were mesophilic, while one study reported a thermophilic reactor with a 200 L working volume, in which 0.8–400 mL/min of air was supplied. The results showed a reduction in H_2S concentration from 3000 to 100 ppm [100]. [101] used microaeration doses of 3.3, 1.8, 3.3, and 2.8 mL/min on days 71, 97, 149, and 168, respectively, in a pilot-scale reactor with a working volume of 247 L, which led to the removal of more than 99 % H_2S. Some other studies also showed about 99 % removal of H_2S using different microaeration rates and intervals [79,102,103].

A study adapted ORP-based controlled microaeration with an airflow rate of 0.14 mL/s while keeping ORP between −320 and −270 mV, which reduced H_2S concentration from 6000 to 30 ppm [79]. Another study used DO as a controlled parameter for microaeration, keeping it in a range of 0.5–2.5 mg/L for the reduction of sludge. The hydrolysis efficacy was improved due to an increase in the abundance of fermentative and hydrolytic bacteria. The proficiency of sludge reduction was enhanced from 42.9 % to 68.3 % as the amount of DO was decreased from 2.5 to 0.5 mg/L in the digester under microaeration [104].

Table 5
Microaeration techniques reported in the literature for the injection of air/oxygen in AD.

Substrate	Microaeration			Objectives	Findings	References
	Technique	Dose	Time			
Swine manure and corn silage	Peristaltic pump	100, 250, 500, 1000- and 2000-mL air/L reactor/d	Intermittent microaeration for 5 min with an air flow rate of 200 mL/min	To enhance VFAs production and investigate microbial dynamics	• VFAs were increased by 20.3 % • Growth of *Clostridium* and *Terrisporobacter* was prominent • *Bifidobacterium* population was significantly increased, leading to the production of LA via Bifidum pathway	[9]
Mixed sludge 40 % PS and 60 % WAS	Syringe	12.5 mL/L·d (Phase I) and 25 mL/L·d (Phase II)	The mentioned dose was injected per day.	To investigate microbial community and methane yield	• Biogas production was increased by 15.8 % • VS removal was enhanced by 18.3 % at microaeration rate (12.5 mL/L·d) • Relative abundance of *Deltaproteobacteria, Anaerolineae, Clostridia, Synergistia* and *Caldilineae* was increased • *Methanosaeta* (67.6 %) was predominated producing methane via acetoclastic pathway	[12]
Buffalo manure	Injection	Phase 1: 7.3 mL/g VS Phase 2: 5.3–9.3 mL/g VS	Phase 1: at 6–47 h Phase 2: at 12, 24 h	To enhance lignin degradation and biogas yield	• Acid-soluble lignin degradation was increased by 78 % • Biogas production was enhanced by 32 % at 7.3 mL O_2/g VS • Aeration dose beyond 7.3 mL O_2/g VS did not improve the lignin degradation.	[16]
Chicken manure	Syringe	7, 10, 20, 30 and 50 mL/g VSin	Microaeration in the first 15 days of digestion	To enhance H_2S removal and methane yield	• Removal of H_2S was enhanced by 58 % • Methane production was increased by 77 % at microaeration (30 mL/g VSin) • *Methanocelleus* was the abundant methanogen	[45]
Mix sludge (PS:60, WAS: 40)	Aeration pipe	0 (raw sludge), 1, 2, 3, 4, 5 and 6 vvm	Microaeration pretreatment for 4 h	To enhance sludge hydrolysis, biogas yield, and microbial community in AD	• Biogas production was increased by 16.4 % • Methane yield was enhanced by 21.6 % • Enrichment of hydrolytic and fermentative bacteria • Methane was produced via acetoclastic methanogenesis	[57]
Corn straw	Syringe	0 (control), 0.05, 0.1, 0.2, 0.4, 0.8, and 1.6 mL/ (g·VS·day	Daily	To investigate microbial community and methane yield	• Methane yield was enhanced by 7.8 % at microaeration (0.2 mL/g·VS·day) • Relative abundance of *Firmicutes, Synergistota* and *Spirochaetota was* increased • *Methanobacterium* was the dominant archaea to produce methane	[46]
Dry corn stover	Mixing in open air	0.1–1 mg/L DO	12-h pretreatment of the substrate at pH 8	To promote corn stover decomposition under real-time pH control	• Methane yield was increased by 16.6 % at pH 8 • Maximum methane concentration was 174.3 mL/g TS with 12 h pretreatment	[77]
Blackwater	Injection	0, 5, 10, 50, and 150 mg g^{-1} COD feed	At the beginning of each cycle	To observe methanogenesis efficiency and microbial communities	• Methanogenesis was enhanced from 20.0 % to 50.9 % • Anaerobic syntrophic bacteria (*Syntrophomonas*) was enriched • *Methanobacterium, Methanolinea,* and *Methanosaeta* were the dominant methanogens at low oxygen dosage • *Ignavibacteriales, Cloacamonales,* and *Rhodocyclales* were the prominent bacteria at high oxygen dosage	[58]
Fresh cow manure	Syringe	0 (i.e., control), 12.5, 37.5 and 62.5 mL/(L·min)	Once a day for 1 min each time	To increase hydrolysis and methane production of high-solids AD of cow manure and observe microbial communities.	• Methane production was increased by 13.1 % • Relative abundance of SOB (*Magnetospirillum*) was increased • SAOB (*Synergistaceae*) was also enhanced • *Methanosaeta* was the dominant acetoclastic methanogen • Cellulase, α-amylase, and triacylglycerol lipase were increased	[98]
Blackwater (toilet wastewater)	Injection	5, 10, 50, and 150 mg O_2/L reactor/cycle	Microaeration for 5 min after 2–5 days	To enhance the hydrolysis of blackwater, methane production and explore microbial communities	• Hydrolysis of blackwater was increased from 34.7 % to 48.7 % • Methane production was improved from 39.6 % to 50.7 % • Growth of hydrolytic and fermentative bacteria was promoted	[99]

(*continued on next page*)

Table 5 (*continued*)

Corn straw	Syringe	0 (control), 0.05, 0.1, 0.2, 0.4, 0.8, and 1.6 mL/ (g·VS·day	Daily	To investigate microbial community and methane yield	• Methane yield was enhanced by 7.8 % at microaeration (0.2 mL/g·VS·day) • Relative abundance of *Firmicutes, Synergistota* and *Spirochaetota was* increased • *Methanobacterium* was the dominant archaea to produce methane	[46]
					• *Methanosarcina* was increased from 1.5 % to 11.4 %	

AD: Anaerobic digestion; PS: Primary sludge; WAS: Waste-activated sludge; COD: Chemical oxygen demand; LA: Lactic acid; vmm: Air volume per gram TS per minute; SOB: Sulfide oxidizing bacteria; SAOB: Syntrophic acetate oxidizing bacteria.

Controlled microaeration with DO from 0.5 to 1.0 mg/L was used in a WWTP with a capacity of 20000 m^3/d for cleaning wastewater (70 % industrial wastewater and 30 % domestic sludge). The pollutants removal efficiency was enhanced for TN (77.2 %), NH_4-N (98.2 %), and COD ($SCOD_{eff}$ of 28.3 mg/L), while decreased sludge production (0.074 g SS/g COD) [106]. In a pilot-scale reactor with a working volume of 200 L, pure O_2 (0.02 %-0.20 %) was injected into the headspace using mass flow controllers, resulting in the production of H_2S-free biogas [67]. In a full-scale mesophilic reactor with a total volume of 338 m, a microaeration dose of 20 m^3/day was applied for the desulfurization of cow manure. This resulted in the reduction of H_2S by 68.2 % due to the enhancement of two SOB (*Halothiobacillus neapolitanus* and *Sulfurimonas denitrificans*) [105]. Similarly, microaeration dosage (0.28–6.0 m^3/h) was applied in full-scale municipal WWTPs in Central Europe, and the results showed an enhancement in desulfurization efficiency from 74 % to 99 % due to increased abundance of SOB. There was also a significant reduction in SCOD from 18 % to 33 % under microaeration [70]. It is concluded that microaeration has been used mainly for H_2S removal in large-scale applications. There is still a lack of microaeration applications on a large scale for enhancing biomethane production and microbial community dynamics. Therefore, microaeration is suggested to be used in large-scale digestors to study and analyze microbial diversity and biomethane production, along with H_2S removal.

3.6. Integrated-microaeration strategies

The efficiency of microaeration could be enhanced by combining it with other approaches (Fig. 6). Microaeration is mostly used with a single type of substrate. The efficiency of microaeration in AD could be enhanced by using a combination of different types of substrates. The advantages of co-digestion include an optimum C/N ratio, maximum nutrient availability, and promoting synergetic interactions between microorganisms [8]. Microaeration, along with co-digestion, could enhance substrate hydrolysis and increase VFAs and biomethane production. Another strategy could be the use of microaeration in biochar-added reactors. Biochar is a kind of carbon-based conducting substance produced as a result of the pyrolysis of biowaste. Biochar has unique characteristics that make it more effective for anaerobic digestion, including sponginess, good electrical conductivity, a definite surface area, cation-exchanging ability, and exterior functional groups [107]. It provides surface area for microorganisms to promote the formation of biofilms, increase interspecies electron transfer, and enhance the stability of AD [7]. The addition of biochar along with microaeration can improve the overall stability of AD for enhanced biomethane yield and H_2S removal from wastewater. Another approach could be microaeration with trace elements in AD. Trace elements (Fe, Zn, Co, Mn, Ni, and Cu) are important parts of coenzymes, cofactors, and prosthetic groups. These elements are crucial for the preservation and development of microbes for metabolic reactions in AD [7]. Microaeration with the addition of trace elements could enhance the growth of hydrolytic and facultative bacteria, which would increase biogas/biomethane production. Another approach could be bioaugmentation with microaeration in AD. Bioaugmentation is the addition of external microorganisms (stress-resistant or highly efficient) to AD. As microaeration enhances hydrolysis, therefore, adding hydrolytic/fermentative bacteria along with microaeration in AD would further improve biogas/biomethane generation. In conclusion, research is needed to implement these strategies in AD to improve the efficiency and stability of the process for achieving maximum biogas/biomethane yield.

3.7. Challenges and future perspectives of microaeration

Application of microaeration should be embedded in AD to improve biowaste conversion rates, biomethane yield, H_2S removal, and process stability. However, further research for industrial-scale applications is required to completely comprehend the fundamental principles and pathways of biomass conversion in AD under microaeration. The following challenges related to microaeration in AD should be addressed for further achievement:

- Microaeration is used in both the liquid and gas phases in full-scale trials for efficient desulfurization of wastewater. However, microaeration in the headspace was often studied, presumably because microaeration in the aqueous phase may produce more nonspecific oxidation of organics, which would result in lower biomethane yield. More research is needed to understand the comparison of microaeration in the headspace and liquid phase and its impact on AD.
- Maintenance of the reactor is required for successful operations of microaeration, particularly for the sporadic elimination of elemental sulfur from the walls of the headspace. Even under aqueous phase microaeration, elemental sulfur is deposited on the headspace walls; however, no information is available on the cleaning process of the headspace of the reactor.
- For industrial applications to be implemented, autonomous regulation of microaeration is a hurdle. Future investigations must use an automated data collection and guidance system for accurate microaeration dosage. Real-time testing can be integrated with the digesters for real-time analysis and assessment of AD to avoid process failure. For instance, ORP sensors can be used in AD for controlled microaeration, while dissolved oxygen (DO) detectors could help maintain a suitable microaeration rate.
- Metabolomics approaches in AD provide significant information about the types of metabolites and the metabolic pathways of bacteria and archaea. The metabolites and pathways helped the researchers in understanding the mechanism of substrate transport across the membrane, metabolism of carbohydrates, proteins, and lipids, quorum sensing, biosynthesis of secondary metabolites, and the osmotic regulation [108]. Metabolomics analysis can be used in AD under microaeration to provide insight into gene expression, metabolite profiles, pathways involved in oxygen stress, and antioxidative mechanisms to optimize AD.
- It is essential to determine the rate of injected oxygen and its usage in different types of AD digesters and operational scenarios under microaeration.

Table 6
Large-scale studies report the effect of various doses of microaeration on AD using different substrates. The schematic diagrams of the reactors used in each study are also shown in the table. All the copyrights have been obtained.

Substrate		Seed sludge	Sludge	Sludge	Raw primary sludge
Reactor configuration	Reactor	Pilot scale CSTR	Pilot-plant scale	Pilot scale CSTR	Pilot anaerobic digester
	Working volume	200 L	200 L	NR	30 L
	Total volume	247 L	250 L	265 L	50 L
	Temperature (°C)	35	35	35	35
Hydraulic retention time (days)		20	20	20–26	20
Microaeration dose		3.3 mL/min (day 71) 1.8 mL/min (day 97) 3.3 mL/min (day 149) 2.8 mL/min (day 168)	0.25 Nm^3/m^3 (day 25) 0.20 Nm^3/m^3 (day 36)	0.16–0.46 NL/L feed (at different intervals depending on hydrogen sulfide)	0.14 mL/s (controlled within the range of ORP from −310 mV and −290 mV)
Effects on AD		Removal of > 99 % of H_2S	Removal of > 99 % of H_2S with a final concentration of about 55 mg/Nm^3	Removal of 99 % of H_2S at 0.46 NL/L_{feed} of oxygen supply when the HS volume was 50.0 and 9.5 L	Reduction in H_2S concentration from 6000 ppm to 30 ppm with a controlled supply of O_2, keeping ORP between −310 mV and −290 mV
Microbial community		NR	NR	NR	NR
References		[101]	[102]	[103]	[79]

Substrate		Sewage sludge	Wastewater	Mesophilic sludge from WWTP	Cow manure
Reactor configuration	Reactor	Pilot scale CSTR	Pilot scale bioreactors	Pilot scale CSTR	Full-scale anaerobic digester
	Working volume	200 L	12.5 L, 25 L	200 L	NR
	Total volume	250 L	NR	250 L	338 m
	Temperature (°C)	35	35	40	35
Hydraulic retention time (days)		19	NR	21	NR
Microaeration dose		0.02 %–0.20 % (v/v) single-point injection of pure O_2 into the HS using mass flow controllers	0.5–2.5 mg/L	0.8–400 mL/min	20 m^3/day
Effects on AD		Production of H_2S-free biogas	Increase in sludge reduction efficiency from 42.9 % to 68.3 % as DO concentration is decreased from 2.5 to 0.5 mg/L	H_2S concentration was decreased from 3000 ppm to < 100 ppm	About 68.2 % reduction in H_2S concentration
Microbial community		NR	Fermentative and predatory bacteria were enriched which increased hydrolysis efficiency and resulted in sludge reduction.	NR	Two SOBs (*Halothiobacillus neapolitanus* and *Sulfurimonas denitrificans*) were the main bacteria involved in bio-desulfurization in the headspace of the digester
References		[67]	[104]	[100]	[105]

Substrate		Municipal wastewater	70 % industrial wastewater and 30 % domestic sewage
Reactor configuration	Reactor	Full-scale WWTPs	Full-scale MSBR
	Working volume	NR	NR
	Total volume	1600–30000 m^3	11035 m
	Temperature (°C)	35	28.7 from July to August 18.2 from October to December
Hydraulic retention time (days)		NR	20
Microaeration dose		0.28–6.0 m^3/h	0.5–1.0 mg/L DO
Effects on AD		Desulfurization efficiency from 74 % to 99 %, reduction in COD	Low production of sludge (0.074 g SS/g COD), efficient removal of pollutants
Microbial community		NR	*Proteobacteria* was the dominant phylum, abundance of denitrifying bacteria involved in denitrification
References		[70]	[106]

ORP: Oxidation-reduction potential; WWTP: Wastewater treatment plant; CSTR: Continuous stirred tank reactor; NR: Not reported; HRT: Hydraulic retention time; Nm^3: Normal meter cube; AD: Anaerobic digestion; mV: Millivolt; H_2S: Hydrogen sulfide; HS: Headspace; SOB: Sulfide oxidizing bacteria; MSBR: Modified sequencing bioreactor.

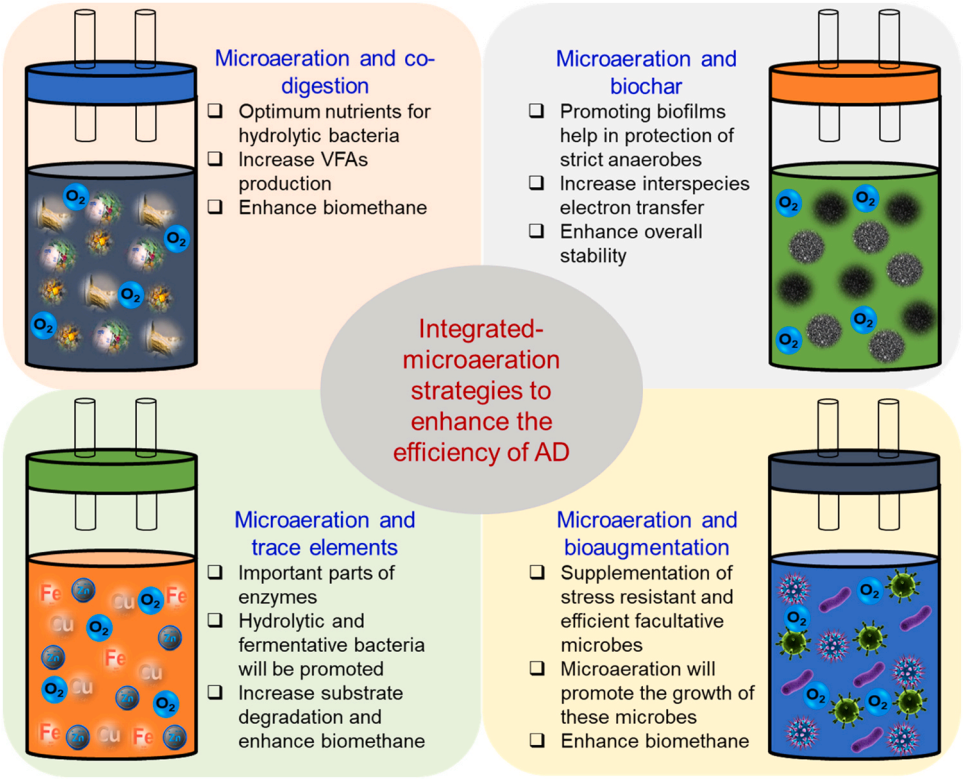


Fig. 6. The integrated-microaeration strategies to enhance the efficiency of the AD system.

- Long-term effects of microaeration on obligate anaerobes and their adaptation to oxygen and ROS need to be investigated.
- It is important to create and validate mathematical models of microaeration in AD using experimental data.

4. Conclusions

Microaeration is preferable in anaerobic digestion (AD) over other approaches owing to its easy operation, eco-friendliness, and cost-effectiveness. It has a positive impact on the AD process, improving the digestibility of the substrate, H_2S removal from wastewater, avoiding high VFAs accumulation, enhancing biomethane content, and maintaining reactor stability. Microbial communities are regulated by a series of enzymatic reactions that establish synergistic relations among various microorganisms. Bacteria (*Firmicutes* and *Bacteroidetes*) and archaeal communities (*Methanosaeta*, *Methanobacterium*, and *Methanosarcina*) have been increased during microaeration in AD. Some methanogens overexpress the genes coding for catalase, superoxidase dismutase, and superoxide reductase, which might protect them from oxygen-toxic effects. Microaeration has a significant role in the desulfurization of wastewater at a large-scale, with successful removal of $\geq$ 90 % H_2S. Meta-omics analysis can help to deeply understand microbial interaction and the enzymatic and biochemical pathways involved in AD under oxygen stress. Further, microaeration could be used in combination with other approaches to enhance the proficiency of the AD system.

CRediT authorship contribution statement

Ubaid Ullah: Visualization, Methodology, Investigation, Data curation, Writing – original draft. **Ali S. Alkorbid**: Visualization, Formal analysis, Validation, Writing – review & editing. **Mohammed Jalalah**: Visualization, Funding acquisition, Formal analysis, Writing – review & editing. **Farid A. Harraz**: Validation, Resources, Funding acquisition, Writing – review & editing. **Adel I. Alalawy**: Visualization, Investigation, Formal analysis, Writing – review & editing. **Sedky H.A. Hassan**: Methodology, Data curation, Formal analysis. **El-Sayed Salama**: Conceptualization, Funding acquisition, Project administration, Supervision, Resources, Validation, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

Acknowledgements

The research team thanks the Deanship of Graduate Studies and Scientific Research at Najran University for supporting the research project through the Nama'a program, with the project code (NU/GP/SERC/13/632–2). This work is supported by the Fundamental Research Funds for the Central Universities (lzujbky-2024-ey12).

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jece.2025.117766.

Data availability

Data will be made available on request.

References

[1] D. Adu, D. Jianguo, S.N. Asomani, A. Abbey, Energy generation and carbon dioxide emission—the role of renewable energy for Green development, Energy Rep. 12 (2024) 1420–1430.
[2] Y. Duan, Z. Wang, P. Ganeshan, T. Sar, S. Xu, K. Rajendran, et al., Anaerobic digestion in global bio-energy production for sustainable bioeconomy: potential and research challenges, Renew. Sustain. Energy Rev. 208 (2025) 114985.

[3] Q. Chen, W. Wu, D. Qi, Y. Ding, Z. Zhao, Review on microaeration-based anaerobic digestion: state of the art, challenges, and prospectives, Sci. Total Environ. 710 (2020) 136388.
[4] Y. Qiu, L. Lower, V. Rondon Berrio, J. Cunniffe, P. Kolar, J. Cheng, et al., Impacts of municipal and industrial organic waste components on the kinetics and potentials of biomethane production via anaerobic digestion, Waste Biomass. Valoriz. (2025).
[5] Andriani, D. Rajani, A. Kusnadi, A. Santosa, A. Saepudin, A. Wresta, et al., A review on biogas purification through hydrogen sulphide removal, IOP Conference Series Earth Environmental Science 483 (2020).
[6] B. Basak, R. Kumar, R.S. Tanpure, A. Mishra, S.K. Tripathy, S. Chakrabortty, et al., Roles of engineered lignocellulolytic microbiota in bioaugmenting lignocellulose biomethanation, Renew. Sustain. Energy Rev. 207 (2025) 114913.
[7] Y. Song, L. Pei, G. Chen, L. Mu, B. Yan, H. Li, et al., Recent advancements in strategies to improve anaerobic digestion of perennial energy grasses for enhanced methane production, Sci. Total Environ. 861 (2023) 160552.
[8] A. Saravanan, P. Senthil Kumar, G. Rangasamy, R. Hariharan, R.V. Hemavathy, P. D. Deepika, et al., Strategies for enhancing the efficacy of anaerobic digestion of food industry wastewater: an insight into bioreactor types, challenges, and future scope, Chemosphere 310 (2023) 136856.
[9] Q. Cao, W. Zhang, T. Lian, S. Wang, F. Yin, T. Zhou, et al., Roles of micro-aeration on enhancing volatile fatty acids and lactic acid production from agricultural wastes, Bioresour. Technol. 347 (2022) 126656.
[10] Li Y., Chen Y., Wu J.J.A.E., Enhancement of Methane Production in Anaerobic Digestion Process: A Review, 240 (2019) 120-137.
[11] Xu Y., Lu Y., Zheng L., Wang Z., Dai X.J.J.O.H.M., Perspective on Enhancing the Anaerobic Digestion of Waste Activated Sludge, 389 (2020) 121847.
[12] Z. Zhou, Q. Ming, Y. An, D. Ruan, G. Chen, H. Wei, et al., Performance and microbial community analysis of anaerobic sludge digestion enhanced by in-situ microaeration, J. Water Process Eng. 42 (2021).
[13] Z. Shao, H. Chen, Z. Zhao, Z. Yang, L. Qiu, X. Guo, Combined effects of liquid digestate recirculation and biochar on methane yield, enzyme activity, and microbial community during semi-continuous anaerobic digestion, Bioresour. Technol. 364 (2022) 128042.
[14] S. Fu, S. Lian, I. Angelidaki, R. Guo, Micro-aeration: an attractive strategy to facilitate anaerobic digestion, Trends Biotechnol. 41 (2023) 714–726.
[15] Y. Zhan, J. Zhu, Y. Xiao, L.C. Schrader, S. Xiao Wu, N. Aka Robinson Jr., et al., Employing micro-aeration in anaerobic digestion of poultry litter and wheat straw: batch kinetics and continuous performance, Bioresour. Technol. 368 (2023) 128351.
[16] I. Zeb, S. Yousaf, M. Ali, A. Yasmeen, A.Z. Khan, J.A. Tariq, et al., In-situ microaeration of anaerobic digester treating buffalo manure for enhanced biogas yield, Renew. Energy 181 (2022) 843–850.
[17] Z. Saboohi, S.E. Hosseini, Advancements in biogas production: process optimization and innovative plant operations, Clean Energy 9 (2025) 52–65.
[18] V. Khanh Nguyen, D. Kumar Chaudhary, R. Hari Dahal, N. Hoang Trinh, J. Kim, S. W. Chang, et al., Review on pretreatment techniques to improve anaerobic digestion of sewage sludge, Fuel 285 (2021).
[19] T. Kunatsa, X. Xia, A review on anaerobic digestion with focus on the role of biomass co-digestion, modelling and optimisation on biogas production and enhancement, Bioresour. Technol. 344 (2022).
[20] R. Bao, Y. Wei, R. Guan, X. Li, X. Lu, S. Rong, et al., High-solids anaerobic co-digestion performances and microbial community dynamics in co-digestion of different mixing ratios with food waste and highland barley straw, Energy 262 (2023) 125529.
[21] Liu W., Sun C., Li W., Li T., Chen Z., Wang J., et al., Sludge Composition and Characteristics Shaped Microbial Community and Further Determined Process Performance: A Study on Full-scale Thermal Hydrolysis-anaerobic Digestion Processes, Journal of Environmental Sciences. 137 (2024) 96-107.
[22] L. Blasco, M. Kahala, S. Ervasti, E. Tampio, Dynamics of microbial community in response to co-feedstock composition in anaerobic digestion, Bioresour. Technol. 364 (2022) 128039.
[23] A.M. Hailu, S.G. Palani, S.L. Asfaw, T.A. Tegaye, Insight into microbial community diversity and composition of two-stage anaerobic digestion: focusing methanogenic stage, Bioresour. Technol. Rep. 15 (2021).
[24] G. Náthia-Neves, M. Berni, G. Dragone, S.I. Mussatto, T. Forster-Carneiro, Anaerobic digestion process: technological aspects and recent developments, Int. J. Environ. Sci. Technol. 15 (2018) 2033–2046.
[25] Y.X. Liew, Y.J. Chan, S. Manickam, M.F. Chong, S. Chong, T.J. Tiong, et al., Enzymatic pretreatment to enhance anaerobic bioconversion of high strength wastewater to biogas: a review, Sci. Total Environ. 713 (2020) 136373.
[26] P.A. Cremonez, J.G. Teleken, T.R. Weiser Meier, H.J. Alves, Two-stage anaerobic digestion in agroindustrial waste treatment: a review, J. Environ. Manag. 281 (2021).
[27] Zhang Y., Li C., Yuan Z., Wang R., Angelidaki I., Zhu G.J.C.E.J., Syntrophy Mechanism, Microbial Population, and Process Optimization for Volatile Fatty Acids Metabolism in Anaerobic Digestion, 452 (2023) 139137.
[28] Kainthola J., Kalamdhad A.S., Goud V.V.J.P.B., A Review on Enhanced Biogas Production from Anaerobic Digestion of Lignocellulosic Biomass by Different Enhancement Techniques, 84 (2019) 81-90.
[29] Zhang M., Qiao Z., Xu S., Luo L., Wong J., Lü F., et al., Pulsed CO2 Feeding Promotes Methane Yield in Anaerobic Digester by Diverting Acidogenic Pathways, (2023) 144756.
[30] Sarker S., Lamb J.J., Hjelme D.R., Lien K.M.J.A.S., A Review of the Role of Critical Parameters in the Design and Operation of Biogas Production Plants, 9 (2019) 1915.
[31] André L., Ndiaye M., Pernier M., Lespinard O., Pauss A., Lamy E., et al., Methane Production Improvement by Modulation of Solid Phase Immersion in Dry Batch Anaerobic Digestion Process: Dynamic of Methanogen Populations, 207 (2016) 353-360.
[32] Xiao L., Liu F., Xu H., Feng D., Liu J., Han G.J.E.C.L., Biochar Promotes Methane Production at High Acetate Concentrations in Anaerobic Soils, 17 (2019) 1347-1352.
[33] Zhao J., Westerholm M., Qiao W., Yin D., Bi S., Jiang M., et al., Impact of temperature and substrate concentration on degradation rates of acetate, propionate and hydrogen and their links to microbial community structure, 256 (2018) 44-52.
[34] R. Conrad, Quantification of methanogenic pathways using stable carbon isotopic signatures: a review and a proposal, Org. Geochem. 36 (2005) 739–752.
[35] G. Li, F. Xu, T. Yang, X. Wang, T. Lyu, Z. Huang, Microbial behavior and influencing factors in the anaerobic digestion of distiller: a comprehensive review, Fermentation 9 (2023).
[36] E. Ripoll, I. López, L. Borzacconi, Hydrogenotrophic activity: a tool to evaluate the kinetics of methanogens, J. Environ. Manag. 270 (2020) 110937.
[37] A. Agrawal, P.K. Chaudhari, P. Ghosh, Anaerobic digestion of fruit and vegetable waste: a critical review of associated challenges, Environ. Sci. Pollut. Res. Int. 30 (2023) 24987–25012.
[38] T. Menzel, P. Neubauer, S. Junne, Role of microbial hydrolysis in anaerobic digestion, Energies 13 (2020).
[39] H.P. Vu, L.N. Nguyen, Q. Wang, H.H. Ngo, Q. Liu, X. Zhang, et al., Hydrogen sulphide management in anaerobic digestion: a critical review on input control, process regulation, and post-treatment, Bioresour. Technol. 346 (2022) 126634.
[40] H. Jung, D. Kim, H. Choi, C. Lee, A review of technologies for in-situ sulfide control in anaerobic digestion, Renew. Sustain. Energy Rev. 157 (2022).
[41] C. Sawatdeenarunat, K.C. Surendra, D. Takara, H. Oechsner, S.K. Khanal.J.B.T., Anaerobic Digestion of Lignocellulosic Biomass: Challenges and Opportunities, 178 (2015) 178186.
[42] D. Nguyen, Z. Wu, S. Shrestha, P.H. Lee, L. Raskin, S.K. Khanal, Intermittent micro-aeration: new strategy to control volatile fatty acid accumulation in high organic loading anaerobic digestion, Water Res. 166 (2019) 115080.
[43] Ward A.J., Hobbs P.J., Holliman P.J., Jones D.L.J.B.T., Optimisation of the Anaerobic Digestion of Agricultural Resources, 99 (2008) 7928-7940.
[44] D. Nguyen, S.K. Khanal, A little breath of fresh air into an anaerobic system: how microaeration facilitates anaerobic digestion process, Biotechnol. Adv. 36 (2018) 1971–1983.
[45] A. Mahdy, Y. Song, A. Salama, W. Qiao, R. Dong, Simultaneous H(2)S mitigation and methanization enhancement of chicken manure through the introduction of the micro-aeration approach, Chemosphere 253 (2020) 126687.
[46] R. Zhu, D.-H. Wang, Y. Zheng, H. Zou, S.-F. Fu, Understanding the mechanisms behind micro-aeration to enhance anaerobic digestion of corn straw, Fuel 318 (2022).
[47] Cheng F., Brewer C.J.R., Reviews S.E., Conversion of Protein-rich Lignocellulosic Wastes to Bio-energy: Review and Recommendations for Hydrolysis+ Fermentation and Anaerobic Digestion, 146 (2021) 111167.
[48] Ariunbaatar J., Panico A., Esposito G., Pirozzi F., Lens P.N.J.A.E., Pretreatment Methods to Enhance Anaerobic Digestion of Organic Solid Waste, 123 (2014) 143-156.
[49] Fu S.-F., Wang F., Shi X.-S., Guo R.-B.J.C.E.J., Impacts of Microaeration on the Anaerobic Digestion of Corn Straw and the Microbial Community Structure, 287 (2016) 523-528.
[50] Duarte M.S., Silva S.R.A., Salvador A.F., Cavaleiro A.J., Stams A.J., Alves M.M., et al., Insight into the Role of Facultative Bacteria Stimulated by Microaeration in Continuous Bioreactors Converting LCFA to Methane, 52 (2018) 6497-6507.
[51] Hanreich A., Schimpf U., Zakrzewski M., Schlüter A., Benndorf D., Heyer R., et al., Metagenome and Metaproteome Analyses of Microbial Communities in Mesophilic Biogas-producing Anaerobic Batch Fermentations Indicate Concerted Plant Carbohydrate Degradation, 36 (2013) 330-338.
[52] Yin J., Yu X., Zhang Y., Shen D., Wang M., Long Y., et al., Enhancement of Acidogenic Fermentation for Volatile Fatty Acid Production from Food Waste: Effect of Redox Potential and Inoculum, 216 (2016) 996-1003.
[53] Goux X., Calusinska M., Fossépré M., Benizri E., Delfosse P.J.B.T., Start-up Phase of an Anaerobic Full-scale Farm Reactor–appearance of Mesophilic Anaerobic Conditions and Establishment of the Methanogenic Microbial Community, 212 (2016) 217-226.
[54] S.-F. Fu, F. Wang, X.-S. Shi, R.-B. Guo, Impacts of microaeration on the anaerobic digestion of corn straw and the microbial community structure, Chem. Eng. J. 287 (2016) 523–528.
[55] J.W. Lim, J.A. Chiam, J.Y. Wang, Microbial community structure reveals how microaeration improves fermentation during anaerobic co-digestion of brown water and food waste, Bioresour. Technol. 171 (2014) 132–138.
[56] Ariesyady H.D., Ito T., Okabe S.J.W.R., Functional Bacterial and Archaeal Community Structures of Major Trophic Groups in A Full-scale Anaerobic Sludge Digester, 41 (2007) 1554-1568.
[57] D. Ruan, Z. Zhou, H. Pang, J. Yao, G. Chen, Z. Qiu, Enhancing methane production of anaerobic sludge digestion by microaeration: enzyme activity stimulation, semi-continuous reactor validation and microbial community analysis, Bioresour. Technol. 289 (2019) 121643.
[58] B. Guo, N. Yu, D.G. Weissbrodt, Y. Liu, Effects of micro-aeration on microbial niches and antimicrobial resistances in blackwater anaerobic digesters, Water Res. 196 (2021) 117035.

[59] J.W. Lim, J.Y. Wang, Enhanced hydrolysis and methane yield by applying microaeration pretreatment to the anaerobic co-digestion of brown water and food waste, Waste Manag. 33 (2013) 813–819.
[60] Tang Y., Shigematsu T., Morimura S., Kida K.J.W.R., The Effects of Micro-aeration on the Phylogenetic Diversity of Microorganisms in A Thermophilic Anaerobic Municipal Solid-waste Digester, 38 (2004) 2537-2550.
[61] Hattori S.J.M., Environments, Syntrophic Acetate-oxidizing Microbes in Methanogenic Environments, 23 (2008) 118-127.
[62] Demirel B., Scherer PJRiES, Bio/Technology, the Roles of Acetotrophic and Hydrogenotrophic Methanogens during Anaerobic Conversion of Biomass to Methane: A Review, 7 (2008) 173-190.
[63] Wang H., Fotidis I.A., Angelidaki I.J.Fme, Ammonia Effect on Hydrogenotrophic Methanogens and Syntrophic Acetate-oxidizing Bacteria, 91 (2015) fiv130.
[64] I. Ramos, M. Fdz-Polanco, The potential of oxygen to improve the stability of anaerobic reactors during unbalanced conditions: results from a pilot-scale digester treating sewage sludge, Bioresour. Technol. 140 (2013) 80–85.
[65] Huertas J.K., Quipuzco L., Hassanein A., Lansing S.J.E., Comparing Hydrogen Sulfide Removal Efficiency in A Field-scale Digester Using Microaeration and Iron Filters, 13 (2020) 4793.
[66] Krayzelova L., Bartacek J., Díaz I., Jeison D., Volcke E.I.P, Jenicek P.J.R.R.E.S., Microaeration for hydrogen sulfide removal during anaerobic treatment: a review, 14 (2015) 703-725.
[67] I. Ramos, R. Perez, M. Fdz-Polanco, The headspace of microaerobic reactors: sulphide-oxidising population and the impact of cleaning on the efficiency of biogas desulphurisation, Bioresour. Technol. 158 (2014) 63–73.
[68] Pokorna D., Zabranska JJBA, Sulfur-Oxidizing Bacteria in Environmental Technology, 33 (2015) 1246-1259.
[69] L. Krayzelova, J. Bartacek, I. Díaz, D. Jeison, E.I.P. Volcke, P. Jenicek, Microaeration for hydrogen sulfide removal during anaerobic treatment: a review, Rev. Environ. Sci. Bio/Technol. 14 (2015) 703–725.
[70] P. Jenicek, J. Horejs, L. Pokorna-Krayzelova, J. Bindzar, J. Bartacek, Simple biogas desulfurization by microaeration - full scale experience, Anaerobe 46 (2017) 41–45.
[71] Song Y., Mahdy A., Hou Z., Lin M., Stinner W., Qiao W., et al., Air Supplement as A Stimulation Approach for the in Situ Desulfurization and Methanization Enhancement of Anaerobic Digestion of Chicken Manure, 34 (2020) 12606-12615.
[72] Khanal S.K., Huang JCJWer, Online Oxygen Control for Sulfide Oxidation in Anaerobic Treatment of High-sulfate Wastewater, 78 (2006) 397-408.
[73] Díaz I., Donoso-Bravo A., Fdz-Polanco MJBT, Effect of Microaerobic Conditions on the Degradation Kinetics of Cellulose, 102 (2011) 10139-10142.
[74] Hirasawa J.S., Sarti A., Del Aguila N.K.S., Varesche MBAJA, Application of Molecular Techniques to Evaluate the Methanogenic Archaea and Anaerobic Bacteria in the Presence of Oxygen with Different COD: Sulfate Ratios in a UASB Reactor, 14 (2008) 209-218.
[75] Chen J.L., Ortiz R., Steele T.W., Stuckey DCJBa, Toxicants Inhibiting Anaerobic Digestion: A Review, 32 (2014) 1523-1534.
[76] Liamleam W., Annachhatre APJBa, Electron Donors for Biological Sulfate Reduction, 25 (2007) 452-463.
[77] H. Xu, Y. Li, D. Hua, Y. Zhao, L. Chen, L. Zhou, et al., Effect of microaerobic microbial pretreatment on anaerobic digestion of a lignocellulosic substrate under controlled ph conditions, Bioresour. Technol. 328 (2021) 124852.
[78] Garcia-Ochoa F., Gomez EJBa, Bioreactor Scale-up and Oxygen Transfer Rate in Microbial Processes: an Overview, 27 (2009) 153-176.
[79] L.D. Nghiem, P. Manassa, M. Dawson, S.K. Fitzgerald, Oxidation reduction potential as a parameter to regulate micro-oxygen injection into anaerobic digester for reducing hydrogen sulphide concentration in biogas, Bioresour. Technol. 173 (2014) 443–447.
[80] Y.-J. Chang, Y.-T. Chang, C.-H. Hung, J.-W. Lee, H.-M. Liao, H.-L. Chou.J.I.B., Microbial Community Analysis of Anaerobic Bio-corrosion in Different ORP profiles, 95 (2014) 9593101.
[81] Morris R.L., Schmidt TMJNRM, Shallow breathing: bacterial life at low O2, 11 (2013) 205-212.
[82] Stolper D.A., Revsbech N.P., Canfield DEJPotNAoS, Aerobic Growth at Nanomolar Oxygen Concentrations, 107 (2010) 18755-18760.
[83] C.G. Liu, C. Xue, Y.H. Lin, F.W. Bai, Redox potential control and applications in microaerobic and anaerobic fermentations, Biotechnol. Adv. 31 (2013) 257–265.
[84] Khanal S.K., Shang C., Huang J.-C.J.W.S., Use of ORP (oxidation-reduction potential) to Control Oxygen Dosing for Online Sulfide Oxidation in Anaerobic Treatment of High Sulfate Wastewater, 47 (2003) 183-189.
[85] Khanal S.K., Huang J.-C.J.J.O.E.E., Anaerobic Treatment of High Sulfate Wastewater with Oxygenation to Control Sulfide Toxicity, 129 (2003) 1104-1111.
[86] Khanal S.K., Huang J.-C.J.W.R., ORP-based Oxygenation for Sulfide Control in Anaerobic Treatment of High-sulfate Wastewater, 37 (2003) 2053-2062.
[87] J. Li, X. Ran, M. Zhou, K. Wang, H. Wang, Y. Wang, Oxidative stress and antioxidant mechanisms of obligate anaerobes involved in biological waste treatment processes: a review, Sci. Total Environ. 838 (2022) 156454.
[88] Fu H., Yuan J., Gao H.J.A.O.B., Microbial Oxidative Stress Response: Novel Insights from Environmental Facultative Anaerobic Bacteria, 584 (2015) 28-35.
[89] Ezraty B., Gennaris A., Barras F., Collet J.-F.J.N.R.M., Oxidative Stress, Protein Damage and Repair in Bacteria, 15 (2017) 385-396.
[90] M. Rezayian, V. Niknam, H.J.Tr Ebrahimzadeh, Oxidative damage and antioxidative system in algae 6 (2019) 1309–1313.
[91] Sies H., Berndt C., Jones D.P., Oxidative Stress, Annu Rev Biochem. 86 (2017) 715-748.
[92] McCord J.M., Keele Jr B.B., Fridovich IJPotNAoS, An Enzyme-based Theory of Obligate Anaerobiosis: the Physiological Function of Superoxide Dismutase, 68 (1971) 1024-1027.
[93] Horne A.J., Lessner DJJFml, Assessment of the oxidant tolerance of Methanosarcina acetivorans, 343 (2013) 13-19.
[94] Brioukhanov A., Netrusov AJAB, Microbiology, Aerotolerance of Strictly Anaerobic Microorganisms and Factors of Defense Against Oxidative Stress: A Review, 43 (2007) 567-582.
[95] Botheju D., Bakke R., Oxygen Effects in Anaerobic Digestion-a Review, (2011).
[96] Lyu Z., Lu YJTIj, Metabolic Shift at the Class Level Sheds Light on Adaptation of Methanogens to Oxidative Environments, 12 (2018) 411-423.
[97] K. Andrade, J. Logemann, K.B. Heidelberg, J.B. Emerson, L.R. Comolli, L.A. Hug, et al., Metagenomic and lipid analyses reveal a diel cycle in a hypersaline microbial ecosystem 9 (2015) 2697–2711.
[98] X. Li, L. Deng, F. Li, D. Zheng, H. Yang, Effect of air mixing on high-solids anaerobic digestion of cow manure: performance and mechanism, Bioresour. Technol. 370 (2023) 128545.
[99] N. Yu, B. Guo, Y. Zhang, L. Zhang, Y. Zhou, Y. Liu, Different micro-aeration rates facilitate production of different end-products from source-diverted blackwater, Water Res. 177 (2020) 115783.
[100] L. Pokorna-Krayzelova, J. Bartacek, S.N. Theuri, C.A. Segura Gonzalez, J. Prochazka, E.I.P. Volcke, et al., Microaeration through a biomembrane for biogas desulfurization: lab-scale and pilot-scale experiences, Environ. Sci. Water Res. Technol. 4 (2018) 1190–1200.
[101] M. Fdz-Polanco, I. Diaz, S.I. Perez, A.C. Lopes, F. Fdz-Polanco, Hydrogen sulphide removal in the anaerobic digestion of sludge by micro-aerobic processes: pilot plant experience, Water Sci. Technol. 60 (2009) 3045–3050.
[102] I. Diaz, A.C. Lopes, S.I. Perez, M. Fdz-Polanco, Performance evaluation of oxygen, air and nitrate for the microaerobic removal of hydrogen sulphide in biogas from sludge digestion, Bioresour. Technol. 101 (2010) 7724–7730.
[103] I. Ramos, I. Diaz, M. Fdz-Polanco, The role of the headspace in hydrogen sulfide removal during microaerobic digestion of sludge, Water Sci. Technol. 66 (2012) 2258–2264.
[104] T. Niu, Z. Zhou, X. Shen, W. Qiao, L.M. Jiang, W. Pan, et al., Effects of dissolved oxygen on performance and microbial community structure in a micro-aerobic hydrolysis sludge in situ reduction process, Water Res. 90 (2016) 369–377.
[105] T. Kobayashi, Y.Y. Li, K. Kubota, H. Harada, T. Maeda, H.Q. Yu, Characterization of sulfide-oxidizing microbial mats developed inside a full-scale anaerobic digester employing biological desulfurization, Appl. Microbiol. Biotechnol. 93 (2012) 847–857.
[106] L.M. Jiang, Z. Zhou, C. Cheng, J. Li, C. Huang, T. Niu, Sludge reduction by a micro-aerobic hydrolysis process: a full-scale application and sludge reduction mechanisms, Bioresour. Technol. 268 (2018) 684–691.
[107] M. Kumar, S. Dutta, S. You, G. Luo, S. Zhang, P.L. Show, , 2021, A Critical Review on Biochar for Enhancing Biogas Production from Anaerobic Digestion of Food Waste and Sludge, 305127143.
[108] Y. Wang, J. Li, M. Liu, L. Gu, L. Xu, J. Li, et al., Enhancement of anaerobic digestion of high salinity food waste by magnetite and potassium ions: digestor performance, microbial and metabolomic analyses, Bioresour. Technol. 388 (2023) 129769.